

Brazil's Marketplace in Motion

A data story of the Olist e-commerce ecosystem (2016–2018)

Dataset: [Brazilian E-Commerce Public Dataset by Olist](#) — 100k real orders placed across Olist's multi-seller marketplace, spanning order status, price, payments, freight, customer & seller geolocation, products, and customer reviews. Sourced directly from Olist's own GitHub repository ([olist/work-at-olist-data](#)), an official mirror of the Kaggle release.

Why this dataset: it is real commercial data (anonymised), genuinely rich (9 linked tables, 45+ usable fields after merging), and varied — numerical (price, freight, weight), categorical (product category, payment type, state), spatial (customer/seller state & zip), and temporal (order timestamps 2016–2018).

Structure of this notebook:

1. Data loading & preliminary EDA
2. 12 analytical questions, each answered with one publication-ready Plotly visualization
3. Summary of key takeaways

```
In [1]: import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
from plotly.subplots import make_subplots

pd.set_option("display.max_columns", 50)
pio.renderers.default = "png" # static, high-res images so the notebook renders identically in Jupyter, HTML
pio.kaleido.scope.default_scale = 2

# ---- CVD-safe palette (Okabe-Ito) -----
GREY      = "#B0B0B0" # muted context
HIGHLIGHT = "#0072B2" # focus blue
ACCENT2    = "#D55E00" # vermillion (secondary highlight)
ACCENT3    = "#009E73" # bluish green
ACCENT4    = "#E69F00" # orange
CVD_SEQ    = ["#0072B2", "#56B4E9", "#009E73", "#E69F00", "#D55E00", "#CC79A7", "#F0E442"]

TEMPLATE = "plotly_white"

def style(fig, title, height=460, width=900):
    fig.update_layout(
        template=TEMPLATE,
        title=dict(text=title, font=dict(size=17, family="Arial", color="#222"), x=0.02, xanchor="left"),
        font=dict(family="Arial", size=13, color="#333"),
        height=height, width=width,
        margin=dict(t=70, l=60, r=40, b=60),
        plot_bgcolor="white", paper_bgcolor="white",
        legend=dict(bgcolor="rgba(0,0,0,0)"),
    )
    fig.update_xaxes(showgrid=False, zeroline=False)
    fig.update_yaxes(showgrid=True, gridcolor="#EDED", zeroline=False)
    return fig

/tmp/ipykernel_621/3179005181.py:10: DeprecationWarning:
Use of plotly.io.kaleido.scope.default_scale is deprecated and support will be removed after September 2025.
Please use plotly.io.defaults.default_scale instead.

pio.kaleido.scope.default_scale = 2
```

1 · Load data

Using the pre-merged master table (see [prepare_data.py](#) in this folder for the full merge/clean pipeline across the 9 raw Olist tables).

```
In [2]: items = pd.read_parquet("../data/master_items.parquet")
delivered = pd.read_parquet("../data/delivered_items.parquet") # delivered orders only (needed for delivery-t.
customers_summary = pd.read_parquet("../data/customers_summary.parquet")

print(f"{items.shape[0]:,} order-items | {items['order_id'].nunique():,} orders | "
      f"{items['customer_unique_id'].nunique():,} unique customers | "
      f"{items['seller_id'].nunique():,} sellers | {items['product_category_name_english'].nunique()} product categories")
items.head(3)
```

112,650 order-items | 98,666 orders | 95,420 unique customers | 3,095 sellers | 72 product categories

	order_id	order_item_id	product_id
0	00010242fe8c5a6d1ba2dd792cb16214	1	4244733e06e7ecb4970a6e2683c13e61 48436dade18ac8b2bce089ec
1	00018f77f2f0320c557190d7a144bdd3	1	e5f2d52b802189ee658865ca93d83a8f dd7ddc04e1b6c2c614352b3
2	000229ec398224ef6ca0657da4fc703e	1	c777355d18b72b67abbef9df44fd0fd 5b51032eddd242adc84c38a

2 · Preliminary Exploratory Data Analysis

A quick look at shape, missingness, and single-variable summaries before moving to the analytical questions (these are exploratory only — not counted among the 12 questions below).

```
In [3]: items.dtypes.value_counts()
```

```
Out[3]: str          16
float64         15
datetime64[us]    7
int64            3
int32            2
bool            2
Name: count, dtype: int64
```

```
In [4]: items.isna().mean().sort_values(ascending=False).head(10).round(3)
```

```
Out[4]: order_delivered_customer_date    0.022
delivery_delay_days                    0.022
delivery_days                          0.022
order_delivered_carrier_date            0.011
review_creation_date                    0.008
review_score                           0.008
product_width_cm                       0.000
product_length_cm                      0.000
product_weight_g                       0.000
product_height_cm                      0.000
dtype: float64
```

```
In [5]: items[['price', 'freight_value', 'item_total', 'review_score']].describe().round(2)
```

```
Out[5]:
```

	price	freight_value	item_total	review_score
count	112650.00	112650.00	112650.00	111708.00
mean	120.65	20.03	140.69	4.03
std	183.63	16.02	190.76	1.39
min	0.85	0.00	6.08	1.00
25%	39.90	13.08	55.23	4.00
50%	74.99	16.27	92.36	5.00
75%	134.90	21.15	157.95	5.00
max	6735.00	409.68	6929.31	5.00

```
In [6]: items['order_status'].value_counts()
```

```
Out[6]: order_status
delivered    110197
shipped      1185
canceled     542
invoiced     359
processing   357
unavailable   7
approved     3
Name: count, dtype: int64
```

```
In [7]: print("Date range:", items['order_purchase_timestamp'].min(), "to", items['order_purchase_timestamp'].max())
```

Date range: 2016-09-04 21:15:19 to 2018-09-03 09:06:57

3 · Analytical Questions

Each question below is multi-dimensional — relating variables, comparing groups, or tracking change over time/space —

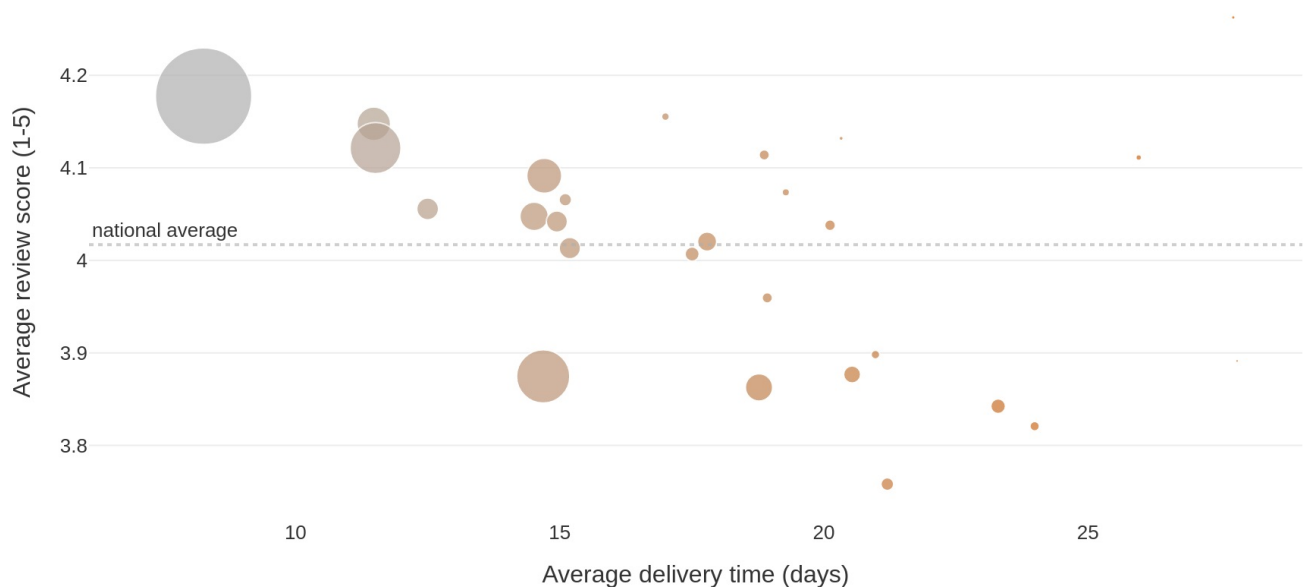
and answered with one dedicated, explanatory Plotly visualization.

Q1 — How does average delivery time vary across Brazilian states, and does slower delivery come with lower customer satisfaction?

```
In [8]: state_perf = (delivered.groupby('customer_state', as_index=False)
    .agg(avg_delivery_days=('delivery_days', 'mean'),
        avg_review=('review_score', 'mean'),
        n_orders=('order_id', 'nunique'))
    .query('n_orders >= 30')
    .sort_values('avg_delivery_days'))

fig = px.scatter(state_perf, x='avg_delivery_days', y='avg_review', size='n_orders',
    color='avg_delivery_days', color_continuous_scale=[GREY, ACCENT2],
    hover_name='customer_state', size_max=45,
    labels={'avg_delivery_days': 'Average delivery time (days)', 'avg_review': 'Average review score'})
fig.update_traces(marker=dict(line=dict(width=1, color='white')))
fig.add_hline(y=state_perf['avg_review'].mean(), line_dash='dot', line_color=GREY, opacity=0.6,
    annotation_text='national average', annotation_position='top left')
fig = style(fig, "Slower-delivery states also rate their orders lower – the North lags on both fronts", height=
fig.update_coloraxes(showscale=False)
fig.show()
```

Slower-delivery states also rate their orders lower — the North lags on both fronts



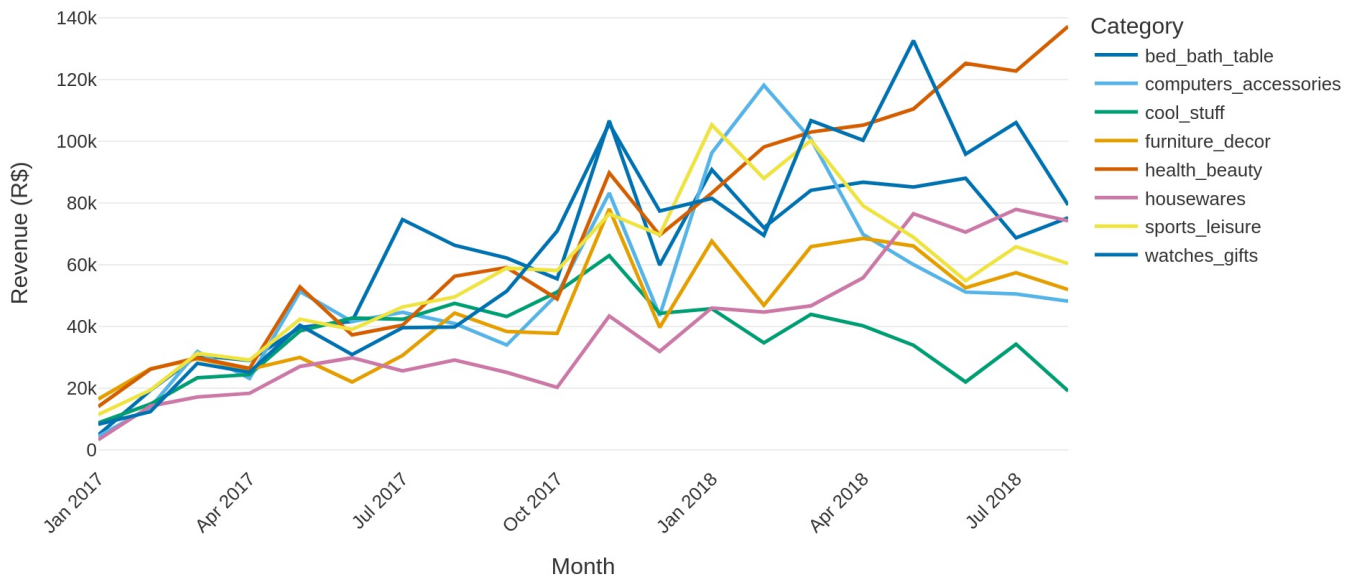
Q2 — Which product categories drive the most revenue, and how has monthly revenue trended over 2017-2018?

```
In [9]: top_cats = (items.groupby('product_category_name_english')['item_total'].sum()
    .sort_values(ascending=False).head(8).index)

monthly_cat = (items[items['product_category_name_english'].isin(top_cats) &
    items['purchase_month'].between('2017-01', '2018-08')]
    .groupby(['purchase_month', 'product_category_name_english'])['item_total']
    .sum().reset_index())

fig = px.line(monthly_cat, x='purchase_month', y='item_total', color='product_category_name_english',
    color_discrete_sequence=CVD_SEQ,
    labels={'purchase_month': 'Month', 'item_total': 'Revenue (R$)', 'product_category_name_english': 'Cat'})
fig.update_traces(line=dict(width=2.5))
fig = style(fig, "Health & beauty and watches/gifts pulled ahead of the pack through 2018", height=480, width=9)
fig.update_xaxes(tickangle=-45)
fig.show()
```

Health & beauty and watches/gifts pulled ahead of the pack through 2018

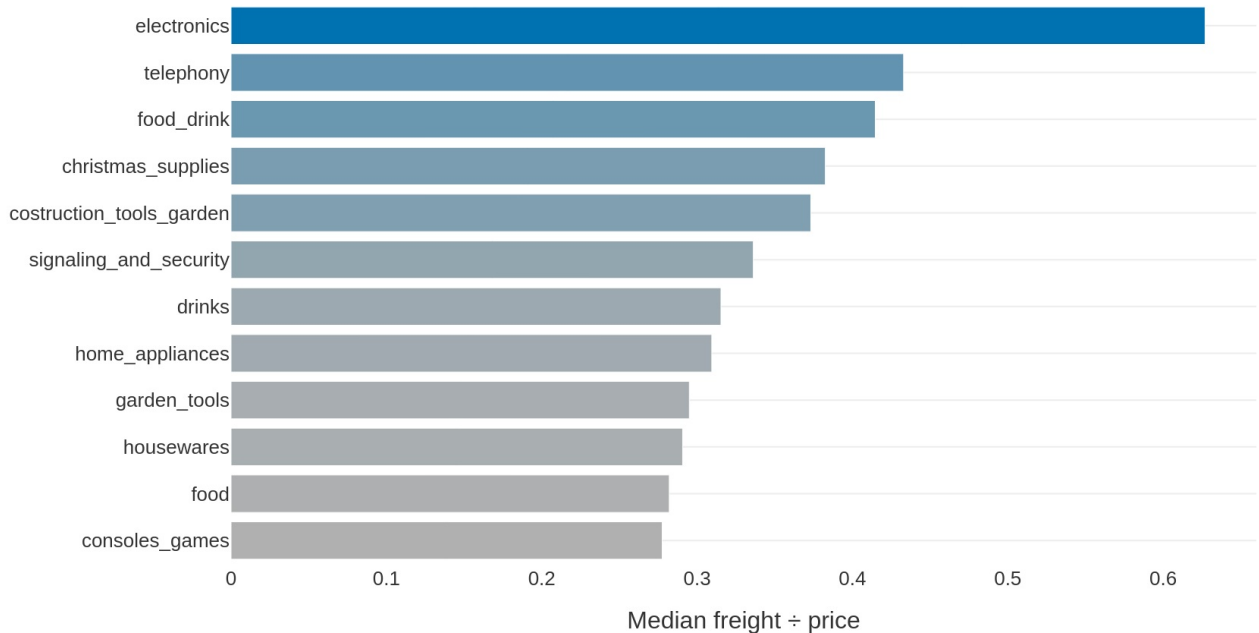


Q3 — How does the freight-to-price ratio differ across product categories, and which categories are costliest to ship relative to their value?

```
In [10]: cat_freight = (delivered.groupby('product_category_name_english')
                        .agg(median_ratio=('freight_ratio', 'median'), n=('order_id', 'count'))
                        .query('n >= 100').sort_values('median_ratio', ascending=False).head(12).reset_index())

fig = px.bar(cat_freight.sort_values('median_ratio'), x='median_ratio', y='product_category_name_english',
             orientation='h', color='median_ratio', color_continuous_scale=[GREY, HIGHLIGHT],
             labels={'median_ratio': 'Median freight ÷ price', 'product_category_name_english': ''})
fig = style(fig, "Bulky, low-value items like furniture & construction goods carry the heaviest relative freight")
fig.update_coloraxes(showscale=False)
fig.show()
```

Bulky, low-value items like furniture & construction goods carry the heaviest relative freight cost



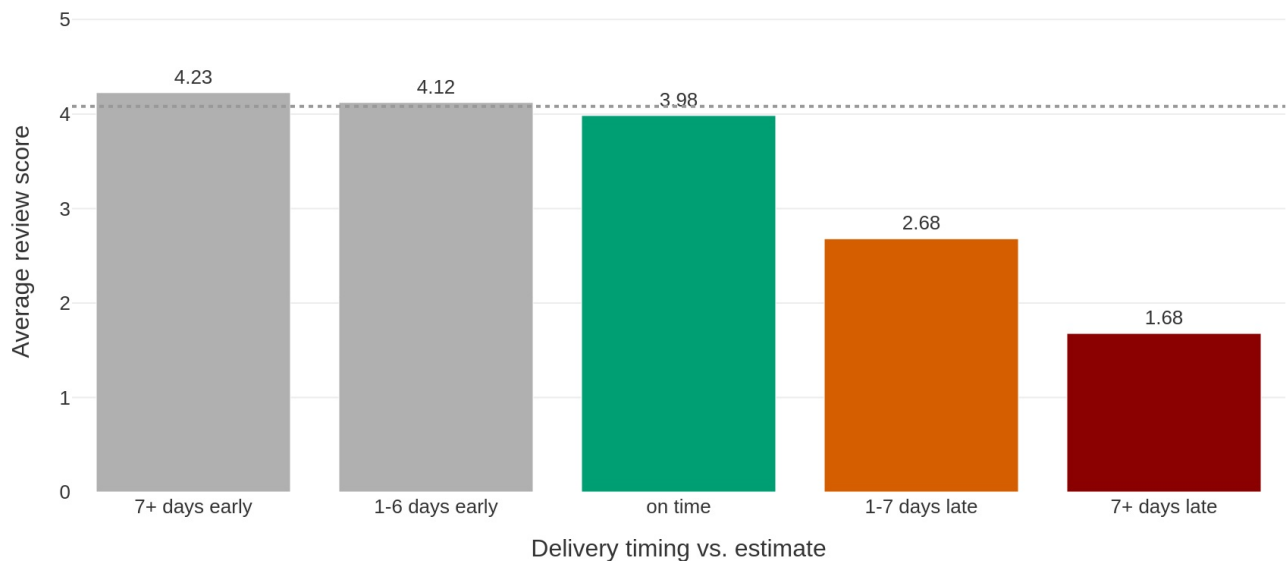
Q4 — Do orders that arrive later than the estimated delivery date get lower review scores?

```
In [11]: d = delivered.dropna(subset=['delivery_delay_days', 'review_score']).copy()
d['delay_bucket'] = pd.cut(d['delivery_delay_days'],
                           bins=[-100, -7, -1, 0, 7, 100],
                           labels=['7+ days early', '1-6 days early', 'on time', '1-7 days late', '7+ days late'])
bucket_rev = d.groupby('delay_bucket', observed=True)['review_score'].mean().reset_index()

colors = [GREY, GREY, ACCENT3, ACCENT2, '#8B0000']
```

```
fig = px.bar(bucket_rev, x='delay_bucket', y='review_score',
             labels={'delay_bucket': 'Delivery timing vs. estimate', 'review_score': 'Average review score'})
fig.update_traces(marker_color=colors, text=bucket_rev['review_score'].round(2), textposition='outside')
fig.add_hline(y=d['review_score'].mean(), line_dash='dot', line_color='#999')
fig = style(fig, "A late delivery costs almost a full star — lateness is the single biggest review-score killer")
fig.update_yaxes(range=[0,5.3])
fig.show()
```

A late delivery costs almost a full star — lateness is the single biggest review-score killer

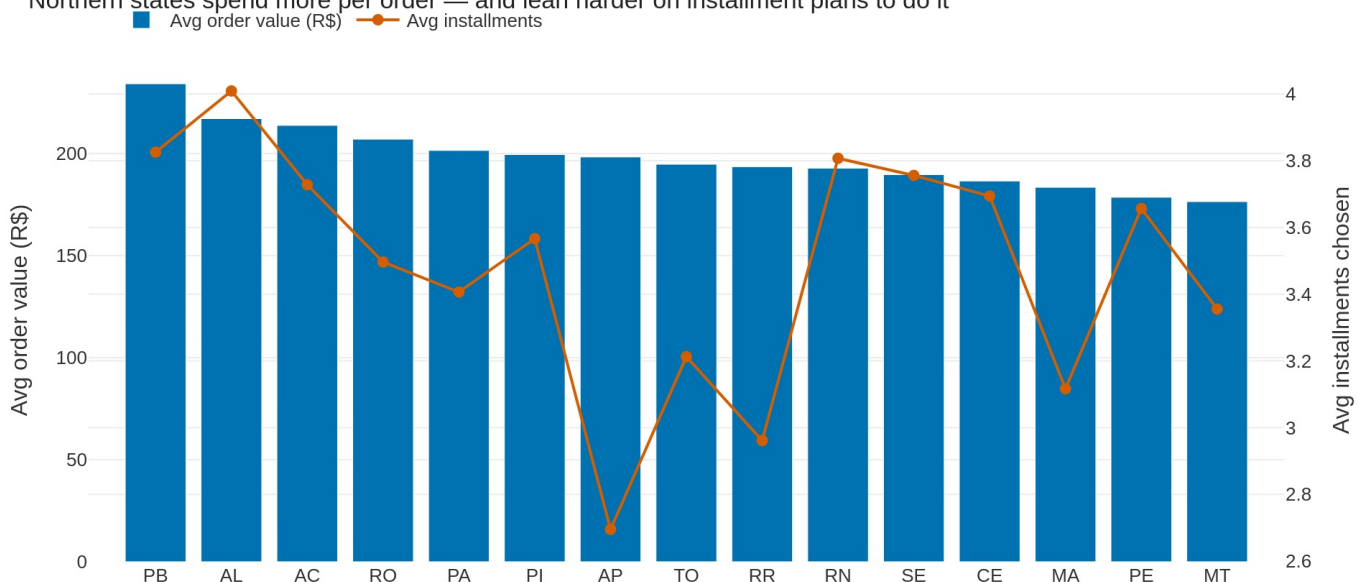


Q5 — Which states have the highest average order value, and does typical installment count vary by region?

```
In [12]: state_value = (items.groupby('customer_state', as_index=False)
                        .agg(avg_order_value=('item_total', 'mean'),
                            avg_installments=('max_installments', 'mean'),
                            n=('order_id', 'nunique'))
                        .query('n >= 30').sort_values('avg_order_value', ascending=False).head(15))

fig = go.Figure()
fig.add_bar(x=state_value['customer_state'], y=state_value['avg_order_value'], name='Avg order value (R$)',
            marker_color=HIGHLIGHT, yaxis='y')
fig.add_scatter(x=state_value['customer_state'], y=state_value['avg_installments'], name='Avg installments',
               mode='lines+markers', marker=dict(color=ACCENT2, size=8), line=dict(width=2, yaxis='y2'))
fig.update_layout(
    yaxis=dict(title='Avg order value (R$)'),
    yaxis2=dict(title='Avg installments chosen', overlaying='y', side='right', showgrid=False),
    legend=dict(orientation='h', y=1.12, x=0.02))
fig = style(fig, "Northern states spend more per order — and lean harder on installment plans to do it", height=
fig.show()
```

Northern states spend more per order — and lean harder on installment plans to do it



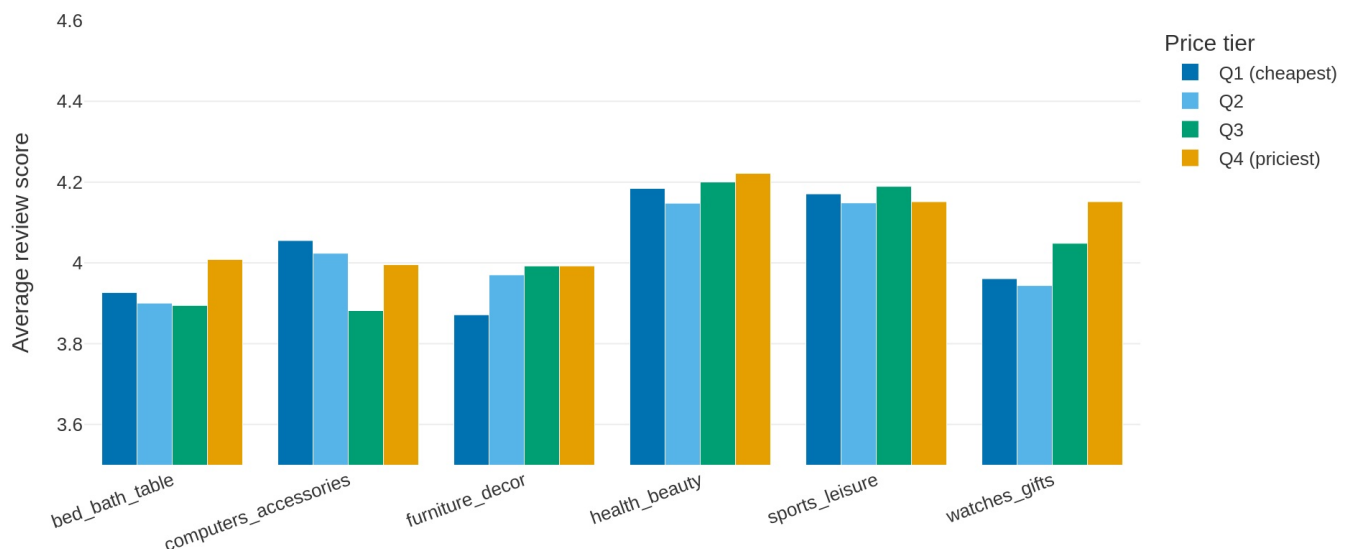
Q6 — How do review scores relate to price tier, and does that relationship hold across product categories?

```
In [13]: d = delivered.dropna(subset=['review_score']).copy()
d['price_tier'] = pd.qcut(d['price'], 4, labels=['Q1 (cheapest)', 'Q2', 'Q3', 'Q4 (priciest)'])
top6 = items.groupby('product_category_name_english')['item_total'].sum().sort_values(ascending=False).head(6)
d6 = d[d['product_category_name_english'].isin(top6)]

pivot = d6.groupby(['product_category_name_english', 'price_tier'], observed=True)['review_score'].mean().reset_index()

fig = px.bar(pivot, x='product_category_name_english', y='review_score', color='price_tier', barmode='group',
             color_discrete_sequence=CVD_SEQ,
             labels={'product_category_name_english': '', 'review_score': 'Average review score', 'price_tier': 'Price tier'})
fig = style(fig, "Higher price tiers don't buy higher satisfaction — the pattern is flat within every top category")
fig.update_yaxes(range=[3.5, 4.6])
fig.update_xaxes(tickangle=-20)
fig.show()
```

Higher price tiers don't buy higher satisfaction — the pattern is flat within every top category

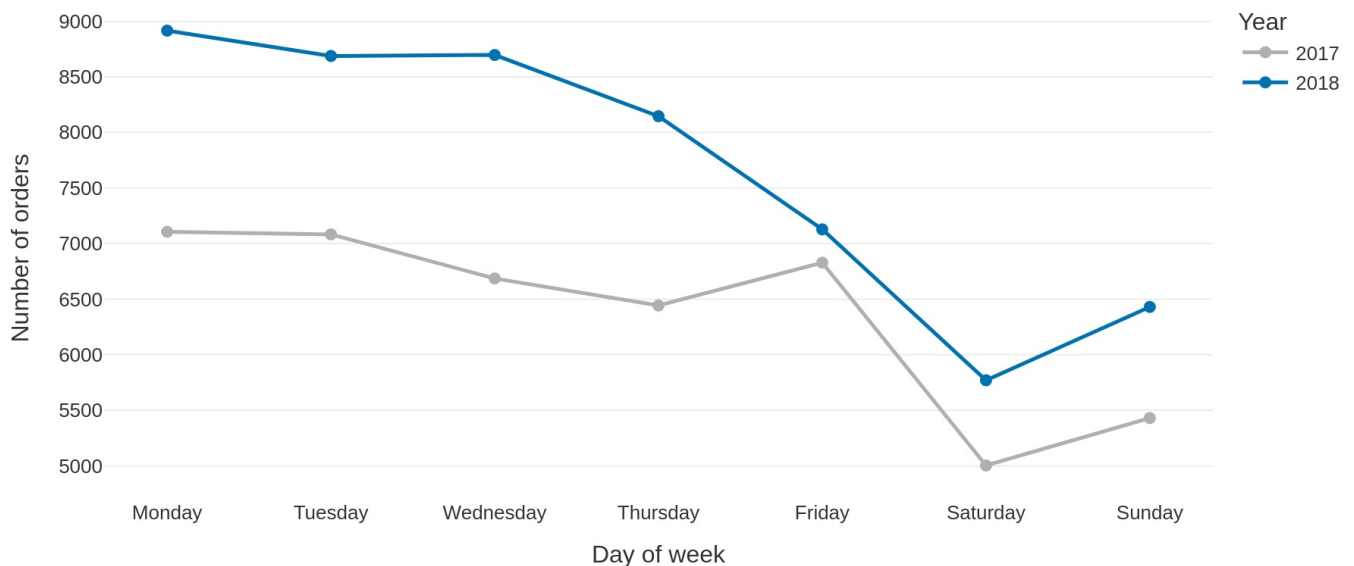


Q7 — What's the weekly rhythm of order placement, and has it shifted between 2017 and 2018?

```
In [14]: dow_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
weekly = (items[items['purchase_year'].isin([2017, 2018])]
         .groupby(['purchase_year', 'purchase_dow'])['order_id'].nunique().reset_index())
weekly['purchase_dow'] = pd.Categorical(w weekly['purchase_dow'], categories=dow_order, ordered=True)
weekly = weekly.sort_values('purchase_dow')

fig = px.line(weekly, x='purchase_dow', y='order_id', color='purchase_year',
             color_discrete_sequence=[GREY, HIGHLIGHT], markers=True,
             labels={'purchase_dow': 'Day of week', 'order_id': 'Number of orders', 'purchase_year': 'Year'})
fig.update_traces(line=dict(width=2.5), marker=dict(size=8))
fig = style(fig, "Weekday shopping habit strengthened in 2018 — weekend orders grew slower than the rest of the")
fig.show()
```

Weekday shopping habit strengthened in 2018 — weekend orders grew slower than the rest of the week

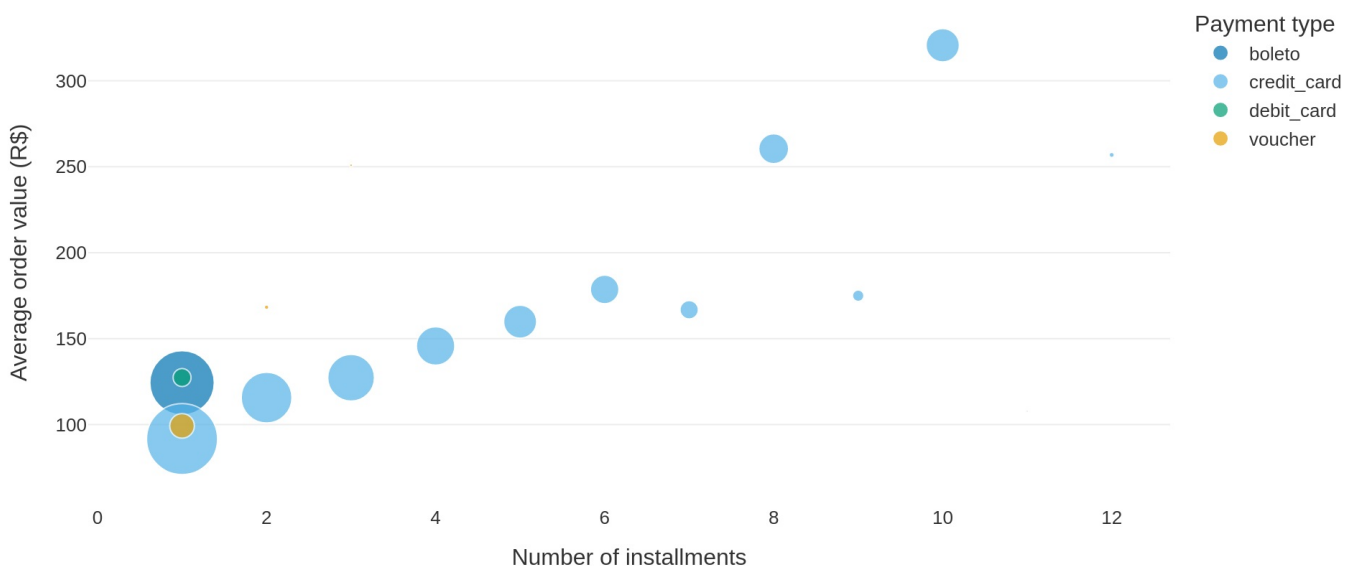


Q8 — Does paying in more installments correlate with higher order value, and does that differ by payment type?

```
In [15]: d = items.dropna(subset=['main_payment_type']).query("main_payment_type != 'not_defined'")
inst = (d.groupby(['main_payment_type', 'max_installments'], as_index=False)
        .agg(avg_value=('item_total', 'mean'), n=('order_id', 'nunique'))
        .query('n >= 20 and max_installments <= 12'))

fig = px.scatter(inst, x='max_installments', y='avg_value', color='main_payment_type', size='n',
                 color_discrete_sequence=CVD_SEQ, size_max=35,
                 labels={'max_installments': 'Number of installments', 'avg_value': 'Average order value (R$)',
                        'main_payment_type': 'Payment type'})
fig = style(fig, "More installments track with bigger baskets — most visibly for credit-card purchases", height=
fig.show()
```

More installments track with bigger baskets — most visibly for credit-card purchases



Q9 — How concentrated is revenue across sellers — do a small number of sellers dominate the marketplace?

```
In [16]: seller_rev = items.groupby('seller_id')['item_total'].sum().sort_values(ascending=False).reset_index()
seller_rev['cum_share'] = seller_rev['item_total'].cumsum() / seller_rev['item_total'].sum()
seller_rev['seller_rank_pct'] = (np.arange(1, len(seller_rev)+1) / len(seller_rev)) * 100

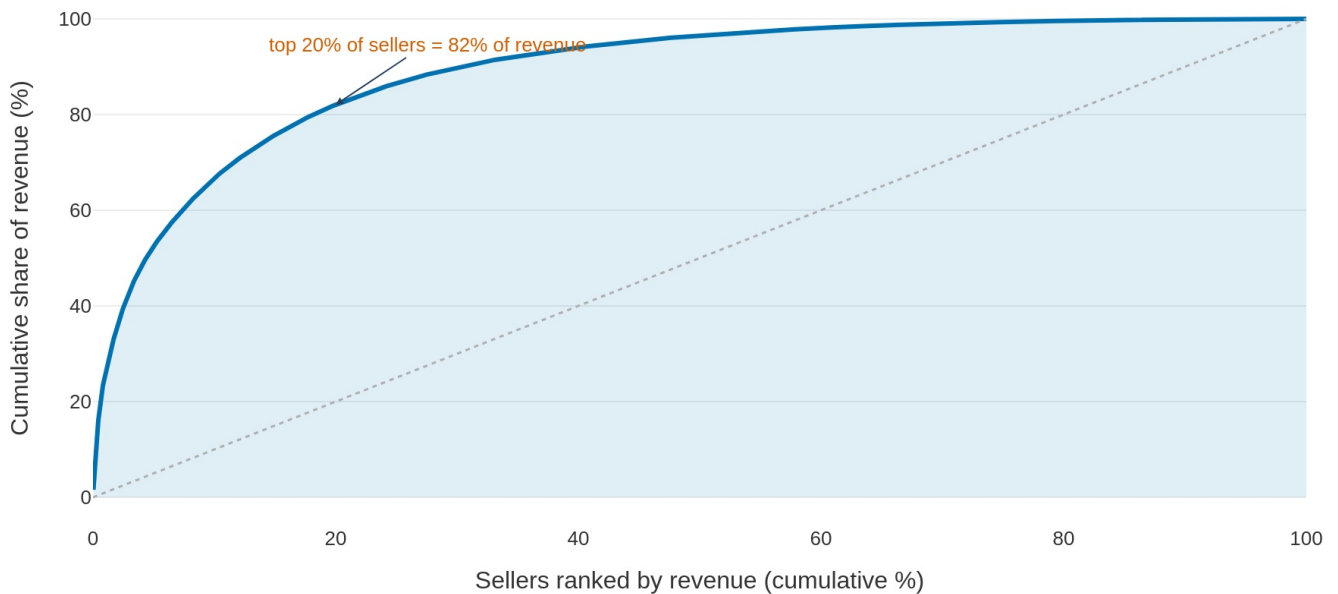
fig = go.Figure()
fig.add_scatter(x=seller_rev['seller_rank_pct'], y=seller_rev['cum_share']*100, mode='lines',
               line=dict(color=HIGHLIGHT, width=3), fill='tozeroy', fillcolor='rgba(0,114,178,0.12)')
fig.add_scatter(x=[0,100], y=[0,100], mode='lines', line=dict(color=GREY, width=1.5, dash='dot'), name='perfect')
```

```

top20_share = seller_rev.loc[seller_rev['seller_rank_pct'] <= 20, 'item_total'].sum() / seller_rev['item_total'].sum()
fig.add_annotation(x=20, y=top20_share*100, text=f"top 20% of sellers = {top20_share:.0%} of revenue",
                  showarrow=True, arrowhead=2, ax=60, ay=-40, font=dict(color=ACCENT2))
fig = style(fig, "Revenue is sharply concentrated: a small slice of sellers accounts for most marketplace revenue")
fig.update_xaxes(title="Sellers ranked by revenue (cumulative %)")
fig.update_yaxes(title="Cumulative share of revenue (%)")
fig.update_layout(showlegend=False)
fig.show()

```

Revenue is sharply concentrated: a small slice of sellers accounts for most marketplace revenue



Q10 — How often are orders shipped cross-state, and does that vary by customer region?

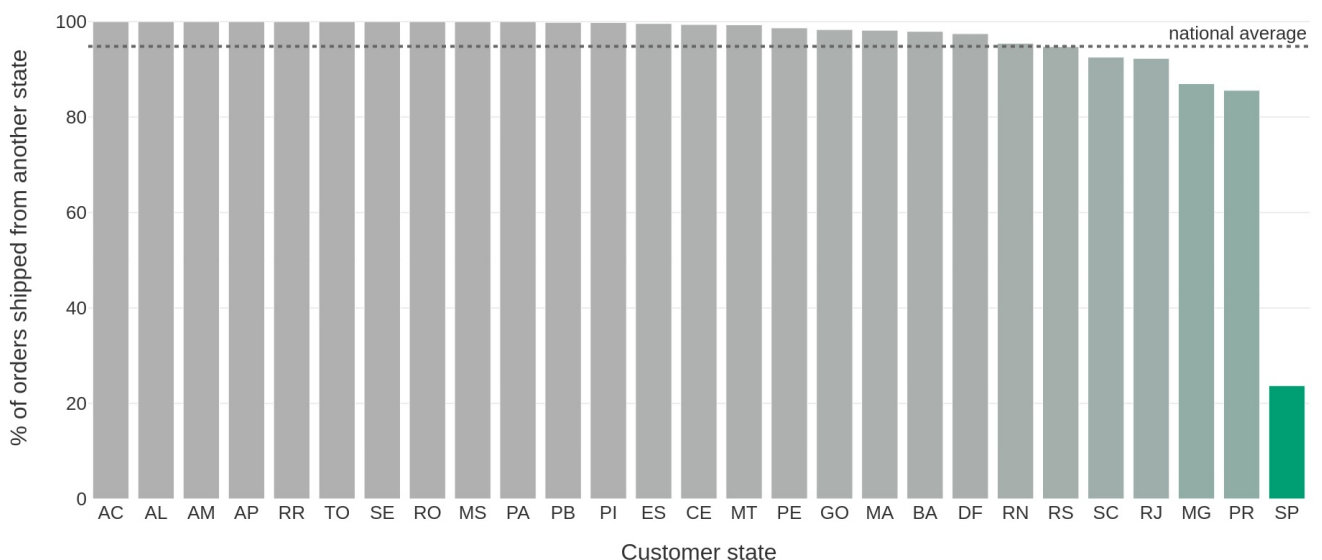
```

In [17]: cross = (items.groupby('customer_state', as_index=False)
                .agg(pct_cross_state=('cross_state_shipment', 'mean'), n=('order_id', 'nunique'))
                .query('n >= 30').sort_values('pct_cross_state', ascending=False))
cross['pct_cross_state'] *= 100

fig = px.bar(cross, x='customer_state', y='pct_cross_state', color='pct_cross_state',
             color_continuous_scale=[ACCENT3, GREY],
             labels={'customer_state': 'Customer state', 'pct_cross_state': '% of orders shipped from another state'})
fig.add_hline(y=cross['pct_cross_state'].mean(), line_dash='dot', line_color='#666',
             annotation_text='national average', annotation_position='top right')
fig = style(fig, "Outside the São Paulo–Rio–Minas core, almost every order ships in from another state", height=100)
fig.update_coloraxes(showscale=False)
fig.show()

```

Outside the São Paulo–Rio–Minas core, almost every order ships in from another state

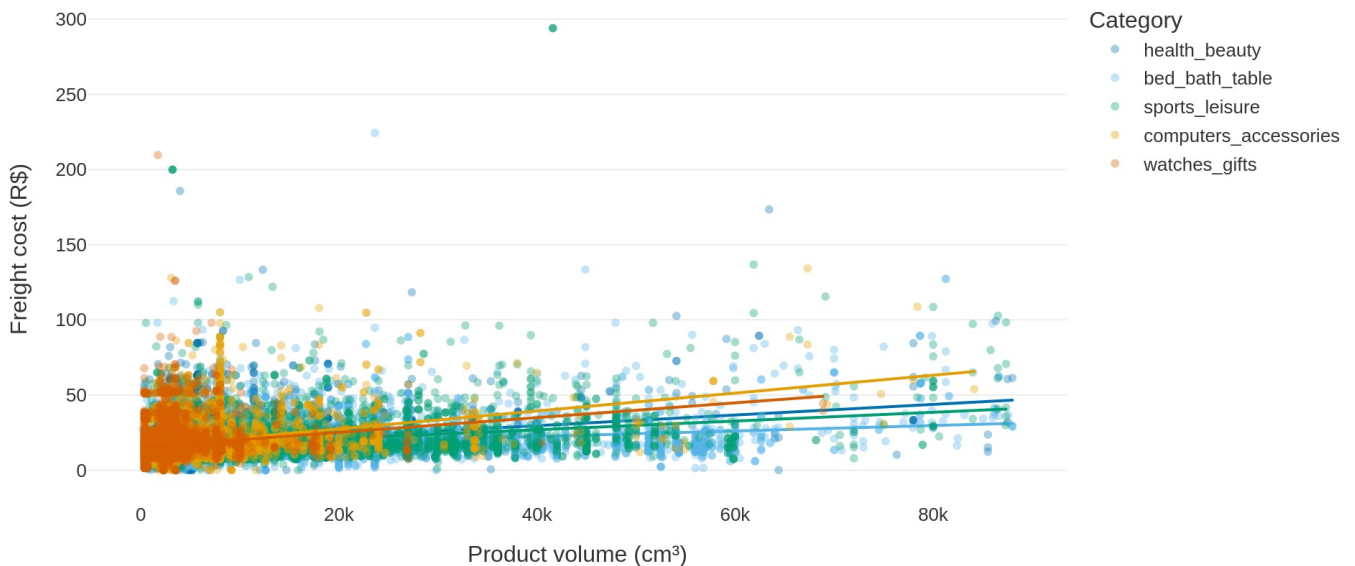


Q11 — How does product size (volume) relate to freight cost, and does this relationship differ for the largest product categories?

```
In [18]: d = delivered.dropna(subset=['product_volume_cm3', 'freight_value']).copy()
d = d[(d['product_volume_cm3'] > 0) & (d['product_volume_cm3'] < d['product_volume_cm3'].quantile(0.98))]
top5 = items.groupby('product_category_name_english')['item_total'].sum().sort_values(ascending=False).head(5)
d5 = d[d['product_category_name_english'].isin(top5)]

fig = px.scatter(d5, x='product_volume_cm3', y='freight_value', color='product_category_name_english',
                trendline='ols', opacity=0.35, color_discrete_sequence=CVD_SEQ,
                labels={'product_volume_cm3': 'Product volume (cm³)', 'freight_value': 'Freight cost (R$)',
                        'product_category_name_english': 'Category'})
fig = style(fig, "Freight scales with product volume — but the slope is steepest for furniture & bed/bath/table")
fig.show()
```

Freight scales with product volume — but the slope is steepest for furniture & bed/bath/table



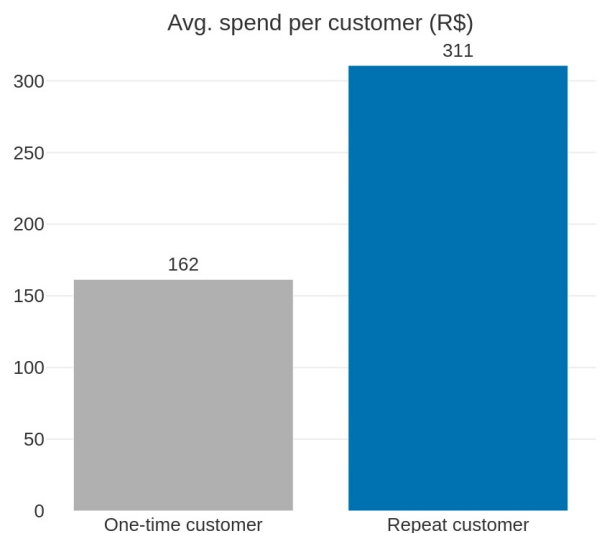
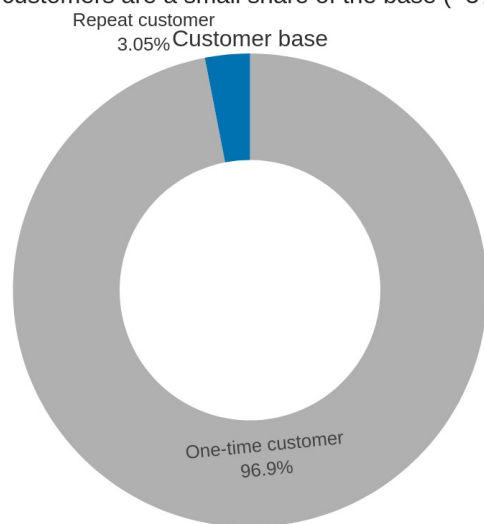
Q12 — What share of customers are repeat buyers, and do repeat customers spend more or rate their experience differently?

```
In [19]: seg = (customers_summary.assign(segment=lambda x: np.where(x['is_repeat'], 'Repeat customer', 'One-time customer')
    .groupby('segment')
    .agg(n_customers=('customer_unique_id', 'count'),
         avg_spent=('total_spent', 'mean'),
         avg_review=('avg_review', 'mean'))
    .reset_index())
seg['pct_of_customers'] = seg['n_customers'] / seg['n_customers'].sum() * 100

fig = make_subplots(rows=1, cols=2, subplot_titles=("Customer base", "Avg. spend per customer (R$)"),
                    specs=[['type': 'domain'], ['type': 'xy']])
fig.add_trace(go.Pie(labels=seg['segment'], values=seg['n_customers'], hole=0.55,
                    marker=dict(colors=[GREY, HIGHLIGHT]), textinfo='percent+label'), row=1, col=1)
fig.add_trace(go.Bar(x=seg['segment'], y=seg['avg_spent'], marker_color=[GREY, HIGHLIGHT],
                    text=seg['avg_spent'].round(0), textposition='outside'), row=1, col=2)
fig = style(fig, "Repeat customers are a small share of the base (~3%) but spend meaningfully more per head", h
fig.update_layout(showlegend=False)
fig.show()

print(seg.round(2))
```

Repeat customers are a small share of the base (~3%) but spend meaningfully more per head



	segment	n_customers	avg_spent	avg_review	pct_of_customers
0	One-time customer	92507	161.53	4.10	96.95
1	Repeat customer	2913	310.75	4.14	3.05

4 · Key takeaways

1. **Delivery speed drives satisfaction more than anything else** — states with longer average delivery times post visibly lower review scores, and orders that arrive even one day late lose the better part of a star versus on-time orders.
2. **Revenue is geographically and structurally concentrated.** São Paulo dominates as both the commercial hub and main shipping origin — most other states rely on cross-state shipments — and the top ~20% of sellers account for the large majority of marketplace revenue.
3. **Price does not buy satisfaction.** Within the best-selling categories, review scores stay essentially flat across price tiers — Olist's satisfaction problem is operational (delivery, freight), not about product pricing.
4. **Freight cost is a size/weight game.** Bulky, lower-value categories like furniture and construction materials carry the worst freight-to-price ratio, and freight cost scales fastest with volume for exactly these categories.
5. **Growth in 2018 was healthy and increasingly weekday-driven,** led by health & beauty and watches/gifts, while installment financing — especially on credit cards — keeps pace with basket size, most strongly in the North.
6. **Repeat purchase is rare (~3% of customers) but valuable** — repeat customers spend more per head, making retention (closely tied to delivery reliability) a clear lever for revenue growth.

(See the accompanying Streamlit dashboard for an interactive exploration of these findings, and the presentation deck for a slide-level summary.)